RESPONSE TO

Request for Information

Accelerating the Adoption and Use of Artificial Intelligence as Part of Clinical Care

RIN 0955-AA13
Docket ID: HHS Health Sector AI RFI

Integrating artificial intelligence into clinical care has implications not only for individual patient outcomes but also for health system performance, equity, and population-level care delivery. Our position is that Artificial Intelligence adoption should be deliberate, not accelerated.

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Submitted to:

Office of the Deputy Secretary
Assistant Secretary for Technology Policy (ASTP)
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Department of Health and Human Services
Re: Request for Information — Accelerating the Adoption and Use of Artificial Intelligence as Part of Clinical Care

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To Whom It May Concern,

We appreciate the opportunity to submit comments in response to the Department of Health and Human Services (HHS) Request for Information (RFI) on accelerating the adoption and use of artificial intelligence (AI) in clinical care. The integration of AI into healthcare represents a significant and timely policy issue with implications for clinical practice, health systems, and patient outcomes. We are approaching this from a *Regulation* perspective.

Our comment reflects an interdisciplinary perspective informed by training and experience across health policy, public health, and clinical medicine. As students engaged in policy development and analysis, and as participants in healthcare delivery and utilization, we bring a multifaceted perspective to the opportunities and challenges of AI implementation in clinical environments. A perspective we believe the agency has overlooked. Our goal is to contribute constructively to ongoing federal efforts by highlighting considerations grounded in clinical realities and systems-level implications.

In the attached comment, we focus on several key themes relevant to responsible AI adoption in healthcare, including the current state of the clinical evidence base, the foundational role of the patient–provider relationship, the role of clinical intuition and experiential dimensions, and the importance of thoughtful implementation pacing. We also outline practical considerations and forward-looking recommendations to support balanced innovation aligned with HHS's mission to enhance the health and well-being of all Americans.

We appreciate HHS's commitment to engaging stakeholders through this RFI and the opportunity to contribute to this important policy dialogue. We hope these observations are helpful as the Department continues to shape strategies that promote innovation while preserving the human-centered foundations of clinical care.

Respectfully submitted,

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Public Comment: Accelerating the Adoption and Use of Artificial Intelligence as Part of Clinical Care

Introduction

We appreciate the opportunity to comment on the Department of Health and Human Services (HHS) Request for Information (RFI), RIN 0955-AA13, *Accelerating the Adoption and Use of Artificial Intelligence as Part of Clinical Care* (U.S. Department of Health and Human Services, 2025b). We are University of Texas at El Paso graduate students engaged in interdisciplinary training and bring unique, firsthand insights as students of policy development, users of AI, healthcare providers, and recipients of care. Brief biosketches of each contributor are provided at the end of this comment.

Policy Context

The mission of the U.S. Department of Health and Human Services is to "enhance the health and well-being of all Americans by providing for effective health and human services and by fostering sound, sustained advances in the sciences underlying medicine, public health, and social services" (U.S. Department of Health and Human Services, 2022). In keeping with this mission, the integration of artificial intelligence (AI) into healthcare is a logical and expected progression. As noted by HHS leadership, AI represents a transformative technology with the potential to reshape health and human services (U.S. Department of Health and Human Services, 2025a). The HHS Artificial Intelligence Strategy, developed in response to OMB Memorandum M-25-21, provides the federal framework for accelerating AI adoption across health and human services and establishes the policy context for the present RFI (Office of Management and Budget, 2025; U.S. Department of Health and Human Services, 2025a).

As AI adoption accelerates, careful consideration of clinical realities will be essential to ensure that innovation translates into meaningful improvements in care delivery. The current AI strategy pays limited attention to the nuanced realities of healthcare delivery that are essential to meaningful, sustained progress. One of the few explicit acknowledgments of these dimensions appears in the statement that "AI-enabled tools will augment clinicians and care teams by supporting decision-making and reducing administrative burden, while preserving clinician autonomy and ensuring that implementation does not compromise the essential human touch needed to build trust and validate the use of these technologies in care delivery." This important observation appears later in the document and is framed more as an implementation consideration rather than as a central guiding principle.

Similarly, the RFI emphasizes operational and systems-level implementation rather than the relational dimensions of care. Within the scope of the RFI, our comment focuses specifically on the research and development framing of AI adoption in healthcare. In response to the RFI, our comment suggests that the adoption of AI in healthcare should proceed deliberately rather than be accelerated, with careful consideration of three interrelated factors: the current evidence base for AI in clinical care, the foundational importance of the patient–provider relationship, and the intuitive and experiential dimensions that shape clinical judgment. These considerations can inform a more balanced, mission-aligned approach to AI adoption that emphasizes thoughtful pacing alongside innovation. The comment concludes by outlining potential directions for the responsible and sustainable integration of AI in healthcare.

Current Evidence Base for AI in Clinical Care

Given the current concentration of evidence within limited and specialty-specific domains, the readiness of AI for broad clinical implementation remains uncertain, particularly in the context of accelerated deployment. As summarized in Table 1, the strength of evidence varies substantially by clinical domain, with the strongest support in narrow, well-defined applications and more limited validation in complex or longitudinal care settings (Kelly et al., 2019; Liu et al., 2019; Nagendran et al., 2020; Topol E.J., 2019b).

Table 1. Evidence strength for selected clinical applications of AI.

Area	Evidence Strength	Clinical Role
Radiology/Pathology	Strong	Assistive
Diabetic Retinopathy	Strong	Screening
ECG/Cardiology	Strong	Detection
Clinical Documentation	Strong	Administrative
Sepsis Alerts	Moderate	Monitoring
Primary Care Diagnosis	Weak	Limited readiness
Oncology Decision Support	Mixed	Advisory

As noted in Table 1, strong evidence for clinical AI is concentrated in interpretive tasks, screening, and administrative automation, and is weakest in primary care diagnostics. This uneven distribution highlights the gap between AI implementation theory and its integration into real-world clinical practice, particularly in settings that rely on contextual judgment. Algorithmic bias further complicates this landscape, as AI systems trained on historical data may reflect underlying inequities and contribute to disparities in performance across populations (World Health Organization, 2021). These considerations may be particularly relevant for populations at heightened risk of inequitable outcomes, including older adults and Veterans with complex care needs. From a policy perspective, this gap raises important questions about AI's readiness for broad clinical implementation, which may outpace the maturity of the current evidence base; accordingly, expanded deployment across clinical settings warrants careful evaluation to ensure effectiveness, safety, and equitable benefit.

Importance of the Patient–Provider Relationship

The April 2025 Office of Management and Budget memorandum outlines a systems-oriented approach to accelerating artificial intelligence adoption across federal operations, emphasizing scale, efficiency, and governance mechanisms to support innovation and public trust (Office of Management and Budget, 2025). Systems approaches prioritize scalability and operational optimization. The HHS OneHHS AI strategy policy (U.S. Department of Health and Human Services, 2025a) reinforces those goals. However, while healthcare systems require system-level optimization, clinical care itself is fundamentally relational, centered on the patient–provider encounter. The HHS implementation framework may underemphasize the fundamental role of the patient–provider relationship and the clinical encounter. This distinction has significant implications for clinical care and policy implementation. The patient–provider relationship has long been recognized within biopsychosocial models of care and is supported by evidence that the therapeutic alliance independently influences clinical outcomes (Engel, 1977; Kelley et al., 2014). Emerging evidence also suggests that AI may negatively affect patients' involvement in decision-making, a core component of patient-centered care (Sauerbrei et al., 2023). Together, these findings reinforce the need for deliberate, phased implementation that preserves relational aspects of care while safeguarding patient-centered decision-making as emerging technologies are integrated into clinical practice.

The Intuitive Dimensions of Clinical Practice

Beyond relational considerations, clinical care also involves cognitive dimensions that are not easily captured by purely analytic frameworks. Clinical decision-making depends not only on formal analytic reasoning but also on intuitive judgment developed through years of experience in complex clinical environments. Intuition in clinical decision-making is often described as tacit knowledge, contextual awareness, and pattern recognition that cannot be fully codified in data-driven models (Croskerry, 2009; Van den Brink et al., 2019). The use of AI in administrative aspects of medicine will free up time, allowing the human connection of listening, empathizing, and building trust to be renewed (Topol E.J., 2019a). As Topol has argued, AI has the potential to restore human-centered aspects of care (Topol E.J., 2019a). Current artificial intelligence systems may not fully capture these embodied and relational forms of judgment. Further, the intuitive processes often serve as early safeguards against diagnostic and therapeutic errors (Croskerry, 2009). AI systems primarily identify statistical associations, whereas clinicians integrate narrative and contextual understanding. Table 2 illustrates key conceptual differences between human clinical encounters and AI. Accelerating reliance on automated outputs in place of clinician reasoning may weaken core mechanisms of the human encounter and healing process. Together, these findings highlight important questions not only about the maturity of the evidence base but also about how emerging technologies are integrated into clinical practice.

Table 2. Conceptual differences between human clinical intuition and AI.

Human Intuition	AI
Context-sensitive	Context-blind
Adaptive	Static
Moral reasoning	Statistical
Meaning-based	Pattern-based
Embodied	Disembodied

The Speed of Implementation and Adoption

The RFI frames AI adoption in terms of acceleration, seeking input on how HHS may "accelerate the adoption and use of artificial intelligence as part of clinical care" (U.S. Department of Health and Human Services, 2025b). This framing assumes that the primary challenge is speed of uptake rather than quality of integration. While evidence maturity is an important consideration, implementation science suggests that the pace at which innovations are introduced into clinical settings is itself a determinant of success or failure. Broadly, evidence-based innovations in healthcare take an estimated 17 to 20 years to reach routine clinical practice, and fewer than half are successfully implemented long term (Bauer & Kirchner, 2020; Morris et al., 2011). While this timeline is often framed as a barrier to innovation, it also reflects the complexity of translating evidence into safe, effective, and sustainable clinical practice. Premature acceleration may compress the very validation processes that ensure patient safety and clinical effectiveness.

Although the evidence base for AI in clinical care is growing, its maturity remains uneven across domains, which argues against broad accelerated deployment. A scoping review of 86 randomized controlled trials evaluating AI in clinical practice found that while 81% reported positive primary endpoints, the majority were single-center studies with limited demographic reporting, raising concerns about generalizability and real-world applicability (Han et al., 2024). Separately, analyses of FDA-authorized AI-enabled medical devices have found that many lack publicly available clinical validation data, and relatively few have been evaluated through randomized controlled trials (Benjamens et al., 2020; Muehlematter et al., 2021). These findings directly inform the RFI's questions regarding the impact of adopted AI tools (Question 10a) and the literature's characterization of the costs and benefits of AI use in clinical care (Question 10b). At present, the literature reflects substantial gaps in the evidence needed to support confident, system-wide deployment.

A comprehensive review of barriers to AI implementation in healthcare identified technological adoption challenges, reliability and validity concerns, and technical limitations as the most frequently cited obstacles, appearing in over 23% of studies (Mehraeen et al., 2025). Similarly, a mixed-methods study combining systematic review evidence with interviews from 69 healthcare professionals found that successful AI implementation requires attention to organizational readiness, stakeholder engagement, and iterative adaptation across phased rollouts (Nair et al., 2024). These findings align with established implementation science frameworks, such as the Consolidated Framework for Implementation Research (CFIR), which emphasize that evidence strength, adaptability, and trialability are critical determinants of whether an innovation is adopted and sustained in practice (Damschroder et al., 2009).

Notably, the implementation science literature distinguishes between adoption and implementation as distinct processes. Adoption refers to the decision to use an innovation, while implementation involves the actions required to integrate it into routine practice (Proctor et al., 2011). Accelerating adoption without adequate attention to implementation—including workforce readiness, workflow integration, and ongoing monitoring—may undermine both clinician trust and patient safety. A recent survey of 43 U.S. health systems found that, while most reported AI adoption across multiple use cases, success varied significantly; for example, only 19% of organizations deploying AI in imaging and radiology reported a high degree of success (Poon et al., 2025). This underscores that adoption without adequate implementation infrastructure does not reliably translate into improved clinical outcomes.

These considerations suggest that the pacing of AI integration should be guided by the maturity of available evidence, the readiness of clinical settings, and the capacity for sustained evaluation. Rather than framing the challenge as one of speed, HHS may benefit from framing the challenge in terms of readiness—asking not how quickly AI can be deployed, but under what conditions deployment is most likely to succeed and sustain.

Recommendation and Future Directions

As healthcare increasingly shifts from treating illness to managing risk, a more deliberate, clinically grounded implementation is warranted to ensure that innovation strengthens—rather than diminishes—the human-centered elements foundational to high-quality care. To support responsible AI integration in clinical care, HHS may consider:

- Aligning AI deployment with evidence maturity across clinical domains
- Incorporating safeguards that preserve relational dimensions of care, including trust, continuity, and shared decision-making
- Incorporating mechanisms to detect and mitigate algorithmic bias to reduce the risk of amplifying health disparities
- Supporting longitudinal evaluation of AI effectiveness and real-world outcomes, especially in domains of chronic illness
- Emphasizing phased implementation over accelerated scaling

In addition to governance considerations, there are domains where AI can be deployed with lower risk and more immediate practical benefits. These include:

- Data management to support the prevention screening of communities and populations
- Non-diagnostic trend detection across laboratory and population data, including automated abnormal-value alerts to support early clinical awareness
- Clinical documentation support, including speech-to-text tools that allow clinicians to focus on patients rather than documentation
- Translation AI for enhancing the patient–provider encounter when language is a barrier

Conclusion

Taken together, these considerations support an AI adoption approach aligned with HHS's core mission of advancing health and well-being while ensuring innovation that strengthens, rather than displaces, the human foundations of clinical care.

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Dr. Petri is a board-certified physiatrist and former U.S. Army physician with senior leadership experience across military, federal, and international health policy, including work with the U.S. Army Surgeon General, Department of Veterans Affairs, and the North Atlantic Treaty Organization. He is currently a PhD student whose work focuses on chronic pain and Veteran-centered care, with interests in integrative medicine, patient–clinician communication, and health policy.

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